



RV College of Engineering®

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NBA Accredited (UG - 6Years)
(PG-3Years)

Department of Electronics & Communication Engineering

Date: 15/04/2026		CIE-I		Max. Marks:60	
Semester: IV		UG		Duration: 02 Hrs	
Course: Microcontroller & Programming (Common to ECE, ETE, EEE)				Code: EI243AI	
Instructions to students: Answer the quiz in the first 2 pages of the answer booklet only					
No	Quiz	M	BT	CO	
1	Mention a typical RISC feature that the ARM Cortex M4 does not follow, which affects code density.	1	2	1	
2	The CoreMark rating of a CPU is 22.72/MHz. What is the rating at 100 MHz?	1	3	3	
3	Which ARM processor family is designed primarily for applications that require high-performance and complex operating systems, like smartphones or tablets?	1	3	1	
4	If instruction in the execution stage of the pipeline has an address indicated by ‘adr’, then what is the address of the instruction in the fetch stage?	1	3	1	
5	Write the use of the program counter and the link register.	1	1	1	
6	Identify the mode of operation of the Cortex M4 core to execute the following assembly code segment. ENTRY MOV R0, #10 MOV R1, #20 ADD R2, R0, R1 STOP B STOP END	1	4	1	
7	Write declarative program statement(s) to store an array of 10 arbitrary words in RAM.	1	2	1	
8	What is the operation of the following ARM instruction? STR r0,[r1], #-4	1	4	1	
9	Give an example of a digital signal controller.	1	2	1	
10	What is the advantage of the THUMB 2 instruction set compared to the ARM ISA.	1	2	1	
TEST					
1.	a. What is a control unit? Describe different ways to generate control signals for PLU operation. b. What is an instruction format? Illustrate with an example.	06 04	2	1	
2.	a. In industrial automation, the separation of stack space between user applications and system applications with separate stack pointers is required to ensure reliable operation. Write the importance of the Main Stack Pointer (MSP) and Process Stack Pointer(PSP). Highlight the status of the SPSEL bit in the control register to differentiate between MSP & PSP access. b. Write the contents in all registers after executing the following code snippet: MOV R0,#0x23 MOV R2,R0 MOV R3,R2 LDR R4,=0x30000000 STR R2,[R4,LSL #2]	05 05	4	4	



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3.	a. Identify the functionality of the following units of Cortex M4 CPUs: i. Debug unit ii. Trace Port iii. Floating point unit iv. Instruction Fetch unit v. Nested Vectored Interrupt Controller b. Write a diagram showing operating modes and states. Clearly show the events that caused the transition of these states and modes.	05	2	3
		05	2	
4.	a. Identify the difference between the following instructions: i. LDRGT r0, [r1, #4]! ii. LDRGT r0, [r1, #4] iii. LDRGT r0, [r1], #4 b. What is address space? Show the differences between Von Neumann and Harvard address spaces with suitable diagrams and examples.	04	4	3
		06	1	
5.	a. A digital controller in an air conditioning system processes sensor data ranging from 0.1°C to 50.0°C. Would you choose fixed-point or floating-point representation? Justify with pros and cons. b. For the code given below, sketch the contents of registers and memory locations with their addresses for little-endian and big-endian processors separately. LDR r0, =X LDR r1, [r0] Area tampdata, data, readonly X DCD 0xaabbccdd	05	3	4
		05	3	1

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	24	00	21	15	07	24	13	16	-	-



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2.	a. R13 is the Stack Pointer(SP) used for accessing the stack memory via PUSH(Storing data on the stack) and POP(Retrieving data from the stack) operations. - Physically, there are two different Stack Pointers: the Main Stack Pointer(MSP) is the default Stack Pointer. It is selected after reset or when the processor is in Handler Mode. - The other Stack Pointer is called the Process Stack Pointer (PSP), normally used when an embedded OS is involved, where the stack for the OS kernel(PSP) and application tasks(MSP) are separated. - The selection of the Stack Pointer is determined by a special register called CONTROL. 02 Marks Control register: <table><tr><td></td><td>31:3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>CONTROL</td><td></td><td></td><td>SPSEL</td><td>nPRIV</td></tr></table> SPSEL (bit 1): Defines the Stack Pointer selection: 0 (default), Thread mode uses MSP. - When this bit is 1, Thread mode uses PSP. - In Handler mode, this bit is always 0, and writing to this bit is ignored. 03 Marks b. Instruction Execution - MOV R0, #0x23 - R0 ← 0x23 - MOV R2, R0 - R2 ← R0 = 0x23 - MOV R3, R2 - R3 ← R2 = 0x23 - LDR R4, =0x30000000 - R4 ← 0x30000000 (loads immediate address) - STR R2, [R4] - Memory[0x30000000] ← R2 = 0x23 05 Marks		31:3	2	1	0	CONTROL			SPSEL	nPRIV	05
	31:3	2	1	0								
CONTROL			SPSEL	nPRIV								
3.	a. Functional Units of Cortex-M4 CPU i. Debug Unit The Debug Unit provides support for debugging and testing programs. It allows breakpoints, watchpoints, single-step execution, and inspection/modification of registers and memory using external tools (like JTAG/SWD debuggers). ii. Trace Port The Trace Port enables real-time instruction and data tracing. It outputs execution information to external tools, helping developers analyze program flow, timing, and performance without stopping the processor.	05										

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iii. Floating Point Unit (FPU)

The **FPU** performs **floating-point arithmetic operations** such as addition, subtraction, multiplication, and division on real numbers. It improves performance in applications like signal processing, control systems, and scientific computations.

iv. Instruction Fetch Unit

The **Instruction Fetch Unit** is responsible for **fetching instructions from memory** and supplying them to the pipeline for execution. It ensures continuous instruction flow and improves execution speed through pipelining.

v. Nested Vectored Interrupt Controller (NVIC)

The **NVIC** manages **interrupt handling** in Cortex-M4. It provides:

Fast interrupt response, Priority-based interrupt handling, Nested interrupt support (higher priority interrupts can preempt lower ones)

5 x 01 Mark

b. Operating states and modes

- The Cortex-M4 processors have two **operation states** and two **modes**.
- In addition, the processors can have **privileged and unprivileged access levels**.

05

Operating states

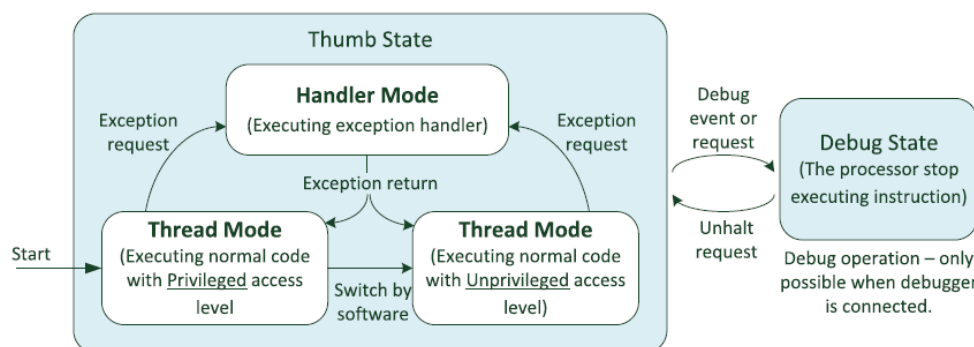
- **Debug state:** When the processor is halted (e.g., by the debugger or after hitting a breakpoint), it enters debug state and stops executing instructions.
- **Thumb state:** If the processor is running program code (Thumb instructions), it is in the Thumb state.

Operation modes (Applicable to Thumb state)

- **Handler mode:** Used to execute an exception handler such as an Interrupt Service Routine (ISR). When in handler mode, the processor always has a privileged access level.
- **Thread mode:** When executing normal application code, the processor can be either in privileged access level or unprivileged access level.

02 Marks

Diagram:



03 Marks

4. a. LDRGT r0, [r1, #4]! → Pre-indexed with write-back

- Address used = $r1 + 4$
- Data loaded from $[r1 + 4]$
- Then r1 is updated to $r1 + 4$

LDRGT r0, [r1, #4] → Pre-indexed without write-back

- Address used = $r1 + 4$

04

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- Data loaded from $[r1 + 4]$
- **r1 is NOT changed**

LDRGT r0, [r1], #4 → Post-indexed

- Address used = r1
- Data loaded from $[r1]$
- Then r1 is updated to $r1 + 4$

04 Marks

b. Address space is the range of memory addresses that a processor can access to store and retrieve data and instructions.

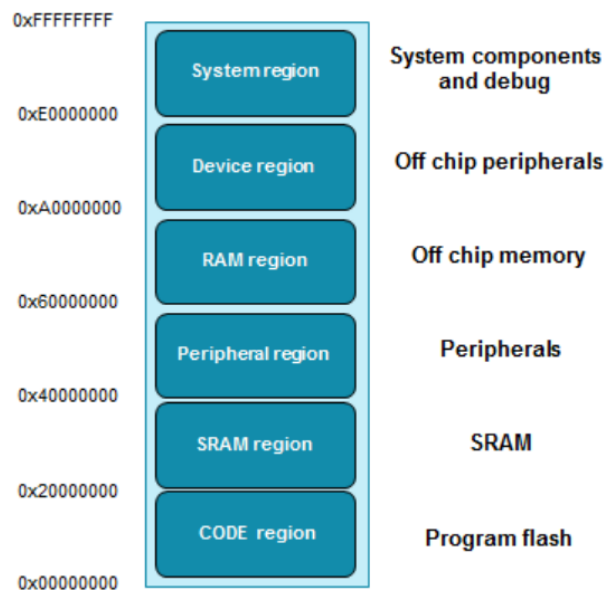
01 Mark

Von Neumann Address Space:

In this architecture, *address space is shared* between program memory & data memory.

E.g: STM32F407VG(based on Cortex M4)

- Total Address space is 4GB(32 bit address)
- The address space is shared between code (Program flash), data(SRAM) IO(peripheral),etc.



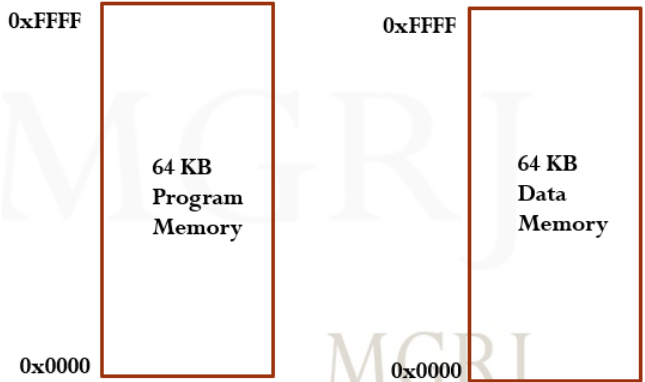
- Single shared bus(Address, data & control: System bus) for Instruction and Data fetching.
- The speed of execution is slower because of the sharing of the bus.
- The complexity of the design of the processor is less because of a single bus.

03 Marks

Harvard Architecture

- In this architecture, *address space is not shared* between program memory & data memory.
- **E.g:8051**
 - Total address space for program memory is 64KB & for data memory is 64KB.

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	- Program memory & data memory locations are <i>separate</i>.  • Separate buses for Instruction and Data fetching. • The speed of execution is faster because separate buses. • The processor design is complex. 02 Marks	
5.	a. For a digital controller in an air conditioning system processing temperature data in the range 0.1°C to 50.0°C, the choice between fixed-point and floating-point representation depends on the required precision, system constraints, and computational resources. Recommended Choice: Fixed-Point Representation 01 Mark Pros of Fixed-Point: • Efficiency: Requires less computational power and memory—ideal for low-cost microcontrollers used in embedded systems. • Faster Execution: Arithmetic operations are simpler and faster than floating-point on microcontrollers without a hardware FPU (Floating Point Unit). • Sufficient Precision: If temperature data is sampled at 0.1°C intervals, fixed-point with a scaling factor (e.g., multiply by 10) can easily represent the range using integers (0 to 500). 02 Marks Cons of Fixed-Point: • Limited Range and Precision Trade-off: Choosing the scaling factor involves a balance between range and precision. • More Programming Effort: Requires careful handling of overflow and rounding manually. 02 Marks	05



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	b.			05
	Component	Little Endian	Big Endian	
	r0	0x1000	0x1000	
	r1	0xAABBCCDD	0xAABBCCDD	
	Memory Order	DD CC BB AA	AA BB CC DD	
05 Marks (with step-by-step analysis)				



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SCRUTINY & EVALUATION OF CIE QUESTION PAPER

Date: 15/04/2026	CIE 1	Max. Marks:60
Semester: IV	UG/PG	Duration: 1 $\frac{1}{2}$ Hrs
Course: Microcontroller & Programming		Code: EI243AI

Declaration by the Course handling faculties:

As a teaching faculty of the course **Microcontroller & Programming** with code: EI243AI,
We hereby confirm that the question paper with Scheme and Solutions has been thoroughly reviewed and we ensure that it adheres to the following criteria:

Sl. No	Criteria	Yes/No
1	The question paper adequately covers the prescribed syllabus contents.	Yes
2	The question paper is in line with the recommended pattern, taking into consideration the structure and format suitable for the Continuous Internal Evaluation.	Yes
3	The question paper is designed to align with the Revised Bloom's Taxonomy, encompassing various levels of cognitive skills such as remembering, understanding, applying, analysing, evaluating, and creating.	Yes
4	The question paper is aligned with the defined course outcomes, ensuring that it effectively assesses the knowledge and skills acquired during the course.	Yes
5	Course handling faculty (as applicable) who are responsible for preparing the question paper, scheme, and Solutions have unanimously agreed to utilize this Question paper for conducting the Continuous Internal Evaluation.	Yes
6	The Question paper, scheme, and complete Solutions have been submitted to the Test coordinators within the designated time frame to ensure the smooth conduction of Continuous Internal Evaluations	Yes

Course handling Faculties:**Name:****Signature**

1. Dr. Govinda Raju M
2. Dr. Roopa J
3. Dr. Ramesh K B
4. Prof. Ravishankar Holla



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To be filled by the Scrutinizer:

Sl. No	Rubrics	Points	
		Max	Awarded
1	Timely submission of the question paper along with the scheme & solution	10	
2	Heterogeneous nature of QP with respect to BTs and Cos	10	
3	Format with proper entry of all particulars, including test, course name, code, date, max marks, BT CO table, and efficient use of paper (proper spacing, figures)	10	
4	No handwritten data or diagrams, and uniform fonts throughout	10	
5	Scheme & complete Solutions in the format	10	
	Total points	50	
Any other comments by the scrutinizer:			

Note: Course coordinators are to obtain the scrutinizer's acceptance by incorporating all suggestions from scrutiny into the final versions of QP, Scheme, and Solutions.

All corrections suggested by the scrutinizer are incorporated and both the copies are re-submitted	Signature of Course coordinator
Accepted/Rejected	Signature of Scrutinizer (Name:)